

WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.2.2 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-23

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core v0.2.x

v0.2.0

- time (a temporal extent) shall be realized as time:interval (unchanged from v.0.1.13)
- Following the OGC Features model, geometry = geolocation = coordinates + elevation
- All temporal<Name> elements are renamed to historical<Name>.
- historical<Name> elements other than historicalLocation shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
historicalTerritory = [{
  "date": "2000-01-01",
  "territory": "<ISO1>"
},
{
  "date": "2024-07-13",
  "territory": "<ISO2>"
}]
```

- "observation" describes time-series (collections) of (individual) observations / measurements.
- "observation"s are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- "deployment" is removed as a concept, most attributes are directly under "observation". - "instrument" links directly to "observation". In alignment with OMS/STA, there is a link to "facility" via "deploymont", but this is not expected to be used at this point.
- "observation"s can be grouped to an "observationSets" (NB: This is a test and not guaranteed to exist in future.)

v0.2.1

- "observation" has been renamed to "observations", because it describes collections (=time-series) of (individual) observations / measurements.
- "observations" are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- The concept of deployment has been developed into "historicalDeployment", which is now a more focussed and intuitive concept. It describes the placement at a specific date of an instrument at a location (=geometry) that can generate observations. It describes the "exposure" of the instrument as well as its "operatingStatus" and "serialNumber" from that date onwards.
- "observations" can be grouped to an "observationSets" (NB: This is experimental and not guaranteed to exist in future.)
- "reporting" and "historicalReporting" are modeled to allow reuse of a reporting template. This is similar to the "schedule" / "historicalObservingSchedule" concept.

v0.2.2

clean up attribute names, consistent singular naming within historical<Name> classes,

WMDR2-core

Version 0.2.1

jkl created on 2026-03-06. Last modified 2026-06-23

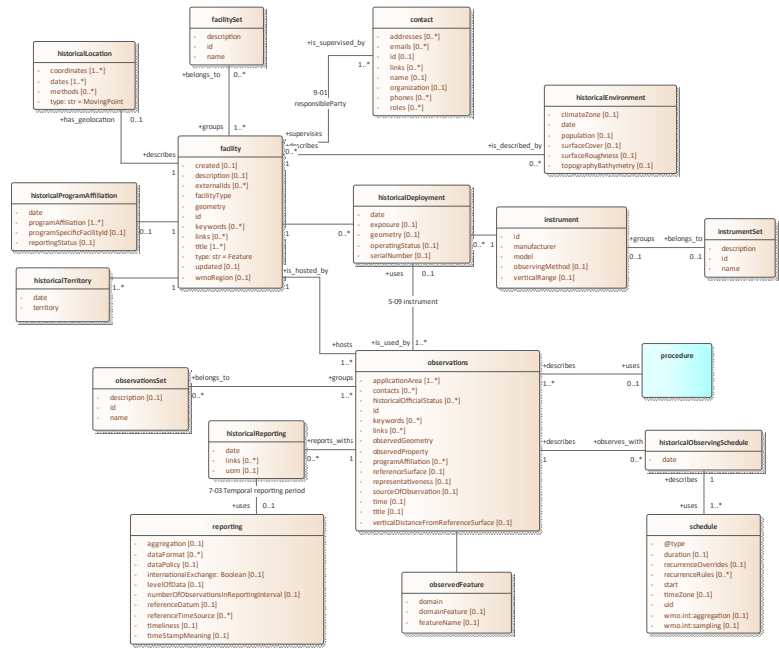


Figure 1: WMDR2-core

WMDR2-core-logical diagram

Class diagram in package 'WMDR2-core'

WMDR2core

Notes

- temporalExtent shall be realized as time:interval
- temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
- Following the OGC Features model, geometry = geolocation = coordinates + elevation

version 0.1.4: simplify observation, specifically specification of 1-01

WMDR2-core-logical

Version 0.1.4

jkl created on 2026-04-13. Last modified 2026-04-13

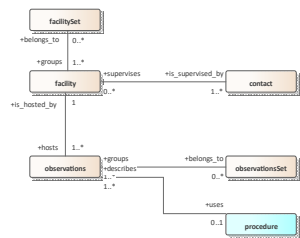


Figure 2: WMDR2-core-logical

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES	
addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact. [Is static True. Containment is Not Specified.]	
name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	
 roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specific contact roles such as owner, pointOfContact, principalInvestigator. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) facility Cardinality: [0..*] Target: Public is_supervised_by (Class) contact Cardinality: [1..*]	

facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Features geometry/time are modeled on fullRecord, not here.

facility

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-23

ATTRIBUTES	
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata. [Is static True. Containment is Not Specified.]	
 description : Private	

ATTRIBUTES	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description. [Is static True. Containment is Not Specified.]	
externalIds : Private Alias: 3-06 (primary & additional) WSIs Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO-WIGOS. [Is static True. Containment is Not Specified.]	
facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value. [Is static True. Containment is Not Specified.]	
geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation [Is static True. Containment is Not Specified.]	
id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:" [Is static True. Containment is Not Specified.]	
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links. [Is static True. Containment is Not Specified.]	
title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName [Is static True. Containment is Not Specified.]	
type : str Private = Feature Facility properties type; const facility.	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalProgramAffiliation Cardinality: [0..1]	
Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public has_geolocation (Class) historicalLocation Cardinality: [0..1]	
Association (direction: Unspecified) Source: Public groups (Class) facility Cardinality: [1..*] Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalTerritory Cardinality: [1..*]	
Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalDeployment Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public is_described_by (Class) historicalEnvironment Cardinality: [0..*]	
Association (direction: Unspecified) 9-01 responsibleParty	

ASSOCIATIONS	
Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]
✍ Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observations Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES	
🔒 description : Private	[Is static True. Containment is Not Specified.]
🔒 id : Private	[Is static True. Containment is Not Specified.]
🔒 name : Private Alias: title	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [1..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]

historicalDeployment

Class in package 'WMDR2-core'

historicalDeployment

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-23

ATTRIBUTES	
🔒 date : Private	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
exposure : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/Exposure	[Is static True. Containment is Not Specified.]
geometry : Private Alias: temporalGeometry Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) geometry is the OGC Records name for geolocation.	[Is static True. Containment is Not Specified.]
operatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) values refer to http://codes.wmo.int/wmdr/InstrumentOperatingStatus	[Is static True. Containment is Not Specified.]
serialNumber : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
Association (direction: Unspecified)	
Source: Public (Class) historicalDeployment Cardinality: [0..*]	Target: Public (Class) instrument Cardinality: [1]
Association (direction: Unspecified) 5-09 instrument	
Source: Public is_used_by (Class) observations Cardinality: [1..*]	Target: Public uses (Class) historicalDeployment Cardinality: [0..1]
Association (direction: Unspecified)	
Source: Public (Class) facility Cardinality: [1]	Target: Public (Class) historicalDeployment Cardinality: [0..*]

historicalEnvironment

Class in package 'WMDR2-core'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

historicalEnvironment

Version 1.0 Phase 1.0 Proposed

ATTRIBUTES	
climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
date : Private [Is static True. Containment is Not Specified.]	
population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility [Is static True. Containment is Not Specified.]	
surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public is_described_by (Class) historicalEnvironment Cardinality: [0..*]	

historicalLocation

Class in package 'WMDR2-core'

historicalLocation

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-22

Alias temporalGeometry

ATTRIBUTES
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Coordinate history array; each entry is lon, lat[, elevation].	[Is static True. Containment is Not Specified.]
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates.	[Is static True. Containment is Not Specified.]
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod	[Is static True. Containment is Not Specified.]
type : str Private = MovingPoint	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public has_geolocation (Class) historicalLocation Cardinality: [0..1]	

historicalObservingSchedule

Class in package 'WMDR2-core'

When are observations made? Describes a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".
Objects like

{"date": date, "schedule": str}, whereby "schedule" is a reference to a re-usable schedule object.

historicalObservingSchedule

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-22

ATTRIBUTES	
date : Private Date from when on the schedule is valid	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified)	

ASSOCIATIONS	
Source: Public describes (Class) historicalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]
✍ Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1]	Target: Public observes_with (Class) historicalObservingSchedule Cardinality: [0..*]

historicalProgramAffiliation

Class in package 'WMDR2-core'

historicalProgramAffiliation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-22

Alias temporalProgramAffiliation

ATTRIBUTES	
🔒 date : Private [Is static True. Containment is Not Specified.]	
🔒 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
🔒 programSpecificFacilityId : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
🔒 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public (Class) facility Cardinality: [1]	Target: Public (Class) historicalProgramAffiliation Cardinality: [0..1]

historicalReporting

Class in package 'WMDR2-core'

ATTRIBUTES	
 date : Private	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval.	[Is static True. Containment is Not Specified.]
 uom : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/unit.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) 7-03 Temporal reporting period	
Source: Public describes (Class) historicalReporting Cardinality: [1]	Target: Public uses (Class) reporting Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1]	Target: Public reports with (Class) historicalReporting Cardinality: [0..*]

historicalTerritory

Class in package 'WMDR2-core'

WMDS reference: 3-02
mandatory for fixed facilities, optional for mobile facilities.
Implemented as [{"date": date, "territory": str}, {}] like other historical<Name> classes.

ATTRIBUTES	
 date : Private	[Is static True. Containment is Not Specified.]
 territory : Private	

ATTRIBUTES	
Alias: temporal Territory	
Values from http://codes.wmo.int/wmdr/TerritoryName	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✎ Association (direction: Unspecified)	
Source: Public (Class) facility Cardinality: [1]	Target: Public (Class) historicalTerritory Cardinality: [1..*]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.
For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-05-22. Last modified 2026-06-13

ATTRIBUTES	
💡 id : Private	[Is static True. Containment is Not Specified.]
💡 manufacturer : Private Alias: 5-09 manufacturer	[Is static True. Containment is Not Specified.]
💡 model : Private Alias: 5-09 model	[Is static True. Containment is Not Specified.]
💡 observingMethod : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial http://codes.wmo.int/wmdr/ObservingMethodEarth	[Is static True. Containment is Not Specified.]
💡 verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public groups (Class) instrument Cardinality: [0..1]	Target: Public belongs_to (Class) instrumentSet Cardinality: [0..1]
✍ Association (direction: Unspecified)	
Source: Public (Class) historicalDeployment Cardinality: [0..*]	Target: Public (Class) instrument Cardinality: [1]

instrumentSet

Class in package 'WMDR2-core'

instrumentSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-22

ATTRIBUTES	
🔒 description : Private	[Is static True. Containment is Not Specified.]
🔒 id : Private	[Is static True. Containment is Not Specified.]
🔒 name : Private Alias: title	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public groups (Class) instrument Cardinality: [0..1]	Target: Public belongs_to (Class) instrumentSet Cardinality: [0..1]

observations

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observations

ATTRIBUTES	
 applicationArea : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Value(s) from http://codes.wmo.int/wmdr/ApplicationArea [Is static True. Containment is Not Specified.]	
 contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation-level contacts. [Is static True. Containment is Not Specified.]	
 historicalOfficialStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) [{"officialStatus": str, "date": date}, {...}], whereby "officialStatus" values refer to http://codes.wmo.int/wmdr/ObservationStatus [Is static True. Containment is Not Specified.]	
 id : Private Observation identifier, e.g. observation:<observedVariable>. [Is static True. Containment is Not Specified.]	
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords. [Is static True. Containment is Not Specified.]	
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links. [Is static True. Containment is Not Specified.]	
 observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry [Is static True. Containment is Not Specified.]	
 observedProperty : Private Alias: OMS.ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in	

ATTRIBUTES	
controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2. Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc. [Is static True. Containment is Not Specified.]	
 programAffiliation : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	Program-affiliation timeline fragments inherited from WMDR1 observations. [Is static True. Containment is Not Specified.]
 referenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	Linked to heightAboveReferenceSurface, from http://codes.wmo.int/wmdr/ReferenceSurfaceType [Is static True. Containment is Not Specified.]
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	http://codes.wmo.int/wmdr/Representativeness [Is static True. Containment is Not Specified.]
 sourceOfObservation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	Value from http://codes.wmo.int/wmdr/SourceOfObservation. If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null. [Is static True. Containment is Not Specified.]
 time : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	Observation temporal extent, date-resolution interval or null. [Is static True. Containment is Not Specified.]
 title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	Human-readable title derived from domain, geometry, observed variable, and optional label. [Is static True. Containment is Not Specified.]

ATTRIBUTES	
verticalDistanceFromReferenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) implement as an array of objects: [{ "value": float, "uom": str }, { ..., ... }], whereby "uom" from http://codes.wmo.int/wmdr/unit [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public groups (Class) observations Cardinality: [1..*] Target: Public belongs_to (Class) observationsSet Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) observations Target: Public (Class) observedFeature	
Association (direction: Unspecified) 5-09 instrument Source: Public is_used_by (Class) observations Cardinality: [1..*] Target: Public uses (Class) historicalDeployment Cardinality: [0..1]	
Association (direction: Unspecified) 3-07 temporalGeometry Source: Public (Class) observations Cardinality: [1] Target: Public (Class) temporalGeometry Cardinality: [0..1] describes has_geolocatino	
Association (direction: Unspecified) Source: Public describes (Class) observations Cardinality: [1] Target: Public reports_with (Class) historicalReporting Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public describes (Class) observations Cardinality: [1..*] Target: Public uses (Class) procedure Cardinality: [0..1] ObservingProcedure(s)	
Association (direction: Unspecified) Source: Public describes (Class) observations Cardinality: [1] Target: Public observes_with (Class) historicalObservingSchedule Cardinality: [0..*]	

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observations Cardinality: [1..*]

observationsSet

Class in package 'WMDR2-core'

grouping of observations (which are themselves time series)

observationsSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-23

ATTRIBUTES	
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Deployment temporal extent, date-resolution interval or null. [Is static True. Containment is Not Specified.]	
 id : Private Deployment identifier; source XML id is preserved when available. [Is static True. Containment is Not Specified.]	
 name : Private [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public groups (Class) observations Cardinality: [1..*]	Target: Public belongs_to (Class) observationsSet Cardinality: [0..*]

observedFeature

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

observedFeature

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-10. Last modified 2026-06-23

ATTRIBUTES

ATTRIBUTES	
domain : Private The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain [Is static True. Containment is Not Specified.]	
domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation. Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like - river basin - lake - river - catchment - canopy - epilimnion - hypolimnion - troposphere - tropopause - stratosphere - thermocline (useful as a general term, but needs "domain") - boundary layer (useful as a general term, but needs "domain") - ... [Is static True. Containment is Not Specified.]	
featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin" [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) observations Target: Public (Class) observedFeature	

procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

ASSOCIATIONS

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1]
ObservingProcedure(s)	

reporting

Class in package 'WMDR2-core'

Reusable reporting template.

reporting

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-23

ATTRIBUTES	
 aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Reporting/aggregation period, renamed from temporalReportingInterval. [Is static True. Containment is Not Specified.]	
 dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat. [Is static True. Containment is Not Specified.]	
 dataPolicy : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/DataPolicy. [Is static True. Containment is Not Specified.]	
 internationalExchange : Boolean Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
 levelOfData : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/LevelOfData. [Is static True. Containment is Not Specified.]	
 numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem. [Is static True. Containment is Not Specified.]	
referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime. [Is static True. Containment is Not Specified.]	
timeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Expected timeliness of reporting, an object with value and uom (default: seconds) [Is static True. Containment is Not Specified.]	
timeStampMeaning : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/TimeStampMeaning. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 7-03 Temporal reporting period Source: Public describes (Class) historicalReporting Cardinality: [1] Target: Public uses (Class) reporting Cardinality: [0..1]	

schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

schedule

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ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
@type : Private JSCalendar object type; const Event. [Is static True. Containment is Not Specified.]	
duration : Private	

ATTRIBUTES	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence rules.	[Is static True. Containment is Not Specified.]
start : Private	
Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.	[Is static True. Containment is Not Specified.]
timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
IANA time zone or UTC default.	[Is static True. Containment is Not Specified.]
uid : Private	
JSCalendar-safe schedule uid.	[Is static True. Containment is Not Specified.]
wmo.int:aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
WMDR-specific aggregation metadata.	[Is static True. Containment is Not Specified.]
wmo.int:sampling : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
WMDR-specific sampling metadata preserved from dataGeneration.sampling.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified)	
Source: Public describes (Class) historicalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 3-07 temporalGeometry Source: Public (Class) observations Cardinality: [1] Target: Public (Class) temporalGeometry Cardinality: [0..1] describes has_geolocatino	